mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA with Circular Positional Encoding and Heteroplasmy Modeling

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Abstract

Mitochondrial DNA (mtDNA) is a 16,569 bp circular genome with central roles in human disease, population genetics, and evolutionary biology. Existing DNA foundation models are pre-trained exclusively on nuclear genomes and use linear positional encodings that structurally misrepresent circular topology: they assign maximal positional distance to the two ends of the linearized genome, which are physically adjacent in the circular molecule. We present **mtDNA-FM**, the first foundation model dedicated to mitochondrial DNA. mtDNA-FM incorporates two biologically motivated architectural design choices: (i) a *circular positional encoding* implemented as a fixed sinusoidal buffer such that positions 0 and 16,568 receive identical encodings, reflecting the physical adjacency of these positions at the D-loop junction in the closed circular molecule and eliminating the artificial positional discontinuity that linear encodings impose; and (ii) a *heteroplasmy projection channel* that accepts a continuous per-position variant allele fraction (0–1) alongside sequence content. Empirical ablation of circular versus standard sinusoidal positional encoding is deferred to the extended paper. We compare mtDNA-FM against DNABERT-2 [Zhou et al., 2024] (117M parameters, BPE tokenization, pre-trained on 32 vertebrate nuclear genomes, 18% of each genome seen) using an identical zero-shot cosine 5-nearest-neighbour protocol on the same train/test split. mtDNA-FM achieves 37.9% on 26-class haplogroup classification versus 66.3% for DNABERT-2 (95% CI 34.4–41.2% vs 63.0–69.5%; intervals do not overlap). Per-class analysis yields four observations: (1) the gap is clade-specific — within mtDNA-FM, all nine European HV haplogroups fall in F1 range 0.000–0.241 and all six African L haplogroups fall in range 0.438–0.857, with no overlap across 15 independent class measurements; (2) the failure is a representation failure — DNABERT-2 correctly classifies H and J using diagnostic positions within 3,000 nt that mtDNA-FM also covers; (3) bidirectional precision and recall collapse for all nine European HV haplogroups confirms embedding-space overlap within the clade; and (4) mtDNA-FM outperforms DNABERT-2 on haplogroups C, F, and E, whose diagnostic positions fall beyond the 3,000-nucleotide truncation limit, providing directional evidence that full-genome coverage is advantageous where truncation removes discriminative information. Pre-trained on 152,590 complete vertebrate mtDNA sequences in two phases (117,615 cross-species, then 34,975 human HmtDB), the model achieves 37.9% on 26-class zero-shot haplogroup classification (cosine 5-NN; 3.85% random baseline, 9.9× lift; 95% CI 34.4–41.2%) and AUROC 0.777 (95% CI 0.731–0.821) on zero-shot pathogenic variant discrimination using ClinVar SNPs against gnomAD common variants. Both evaluations use the Phase 1 checkpoint; Phase 1 vs. Phase 2 ablation is deferred to the extended paper. Pre-trained weights, a Gradio demonstration interface, and the full DVC pipeline are publicly available.

Availability: `https://github.com/vthawfeek/mtdna-foundation-model`, `https://huggingface.co/vthawfeek/mtdna-foundation-model`, `https://huggingface.co/spaces/vthawfeek/mtdna-fm-demo`

1 Introduction

Mitochondrial DNA (mtDNA) is a 16,569 bp circular, double-stranded genome present in hundreds to thousands of copies per cell [Anderson et al., 1981]. Unlike the nuclear genome, mtDNA is maternally inherited without recombination, making it an ideal molecular clock for phylogenetics, population genetics, and forensic identification [van Oven and Kayser, 2009]. Clinically, pathogenic mtDNA variants cause severe multi-system disorders including MELAS, LHON, and Leigh syndrome, collectively affecting at least 1 in 5,000 individuals [DiMauro and Schon, 2003]. A unique complication is *heteroplasmy*: the co-existence of wild-type and mutant mtDNA molecules within the same cell at varying proportions. Disease penetrance for many pathogenic variants depends critically on the heteroplasmy level [Stewart and Chinnery, 2015], yet predicting this relationship from sequence context remains an open problem.

DNABERT-2 [Zhou et al., 2024], HyenaDNA [Nguyen et al., 2023], the Nucleotide Transformer [Dalla-Torre et al., 2023], and Evo [Nguyen et al., 2024] show that pre-training on DNA sequences yields representations that transfer across genomic tasks. But all four are pre-trained exclusively on nuclear genomes and share two design gaps for mtDNA:

1. **Linear positional encoding.** Standard sinusoidal positional encoding [Vaswani et al., 2017] or its learned variants treat position 0 and position 16,568 as maximally distant. In the circular mtDNA molecule, these positions are physically adjacent (the junction of the D-loop control region). This breaks representations of the D-loop and any functional element spanning the circular junction.

2. **No heteroplasmy channel.** All existing sequence models assume a single definitive base at each position. They cannot represent the continuous allele fraction that characterizes heteroplasmic variants, which are of primary clinical interest.

Nuclear DNA also contains repeat elements, splice sites, CpG islands, and regulatory grammar absent from mtDNA. Representations built from billions of bases of nuclear sequence may not transfer to a 16,569 bp genome with different base composition, gene density, and evolutionary constraint.

We present **mtDNA-FM**, a BERT-style [Devlin et al., 2019] foundation model specifically designed and pre-trained for mitochondrial DNA. Our contributions are:

1. A **circular positional encoding** that encodes the biologically correct circular topology as a non-learnable buffer: mtDNA is a closed circular molecule, and positions 0 and 16,568 are physically adjacent at the D-loop control region boundary. Standard sinusoidal encodings assign maximal positional distance to these adjacent positions, creating a topologically incorrect inductive bias. Our circular encoding eliminates this mismatch, ensuring smooth representations at the genome junction. Empirical ablation against sinusoidal PE is deferred to the extended paper (Section 3.1).

2. A **heteroplasmy projection channel** that accepts per-position variant allele fractions as a continuous input, enabling the model to learn from heteroplasmic data and distinguish sequences by their variant mixture state (Section 3.1).

3. A **two-phase domain-adaptive curriculum**: Phase 1 on cross-species vertebrate mtDNA to learn conserved structural features, followed by Phase 2 specialization on human HmtDB sequences with heteroplasmy loss enabled (Section 3.2).

4. **Zero-shot evaluation** on 26-class haplogroup classification and pathogenic variant discrimination, with per-class analysis against DNABERT-2 (Section 4).

2 Related Work

2.1 Genomic Foundation Models

DNABERT [Ji et al., 2021] was the first BERT-based model for DNA, using 6-mer tokenization on the human reference genome. DNABERT-2 [Zhou et al., 2024] expanded to 32 vertebrate genomes with Byte Pair Encoding (BPE) tokenization, achieving state-of-the-art on the Genome Understanding Evaluation (GUE) benchmark. The Nucleotide Transformer [Dalla-Torre et al., 2023] scales to 2.5B parameters across 850+ genomes. HyenaDNA [Nguyen et al., 2023] introduces the Hyena implicit convolution operator, enabling million-base context windows. Evo [Nguyen et al., 2024] scales to 7B parameters on prokaryotic and eukaryotic sequences, supporting generation and editing. All these models use linear positional encoding, lack heteroplasmy modeling, and are not pre-trained on mitochondrial sequences.

2.2 mtDNA Bioinformatics Tools

Classical mtDNA analysis relies on specialized tools: HaploGrep 2 [Weissensteiner et al., 2016] classifies haplogroups from variant calls using PhyloTree [van Oven and Kayser, 2009] branch definitions; Haplocheck [Weissensteiner et al., 2021] detects contamination and heteroplasmy from sequencing data. Pathogenicity databases include MITOMAP [Lott et al., 2013], ClinVar [Landrum et al., 2018], and HelixMTdb [Bolze et al., 2022]. MitImpact [Castellana and Mazza, 2013] aggregates computational pathogenicity scores for protein-coding variants. None of these tools use pre-trained deep learning representations.

2.3 Variant Effect Prediction

Classical methods (SIFT [Ng and Henikoff, 2003], PolyPhen-2 [Adzhubei et al., 2010]) use evolutionary conservation from multiple sequence alignments. Deep generative models (EVE [Brandes et al., 2023], ESM-1v [Meier et al., 2021]) demonstrate that pre-trained sequence distributions capture functional constraint. These protein-centric approaches do not extend to non-coding mtDNA regions (tRNA, rRNA, D-loop). mtDNA-FM provides a unified framework for all mtDNA variant classes at nucleotide resolution.

2.4 Positional Encoding Variants

Standard sinusoidal PE [Vaswani et al., 2017] and its learnable counterpart treat position as a linear index. Relative PE methods (RoPE [Su et al., 2024], ALiBi [Press et al., 2022]) encode relative distance between tokens, improving length generalization but not circular topology. To our knowledge, no prior sequence model uses circular positional encoding; our formulation is the first application to biological sequences.

3 Methods

3.1 Architecture

mtDNA-FM adopts a BERT-style [Devlin et al., 2019] transformer encoder with 6 layers, 8 attention heads, 256-dimensional hidden states, and 1,024-dimensional feed-forward layers ($\sim$5.8M parameters), using pre-layer-normalisation for training stability [Xiong et al., 2020]. Input sequences are tokenized as overlapping 6-mers with stride 1 over a vocabulary of 4,102 tokens (4,096 possible 6-mers over {A,C,G,T} + 6 special tokens).

3.1.1 Circular Positional Encoding

Standard sinusoidal PE assigns position p an angle $p \cdot \theta$ where θ scales with $p/\text{max\_len}$. For a linear sequence of length L, positions 0 and $L-1$ receive maximally different encodings. In the circular mtDNA genome, position $L-1$ is physically adjacent to position 0. We resolve this by substituting a circular angle:

$$\phi_p = \frac{2\pi p}{L_{\text{genome}}} \tag{1}$$

where $L_{\text{genome}} = 16,569$ bp is the rCRS genome length. The positional encoding is then:

$$\text{PE}(p, 2i) = \sin\left(\frac{\phi_p}{10000^{2i/d}}\right) \tag{2}$$

$$\text{PE}(p, 2i+1) = \cos\left(\frac{\phi_p}{10000^{2i/d}}\right) \tag{3}$$

where $d = 256$ is the hidden dimension. At $p = L_{\text{genome}}$, $\phi_p = 2\pi$, so $\sin(2\pi) = \sin(0) = 0$ and $\cos(2\pi) = \cos(0) = 1$. The circular junction is thus mathematically smooth. The encoding is registered as a non-learnable `nn.Buffer`, ensuring it cannot be modified during training.

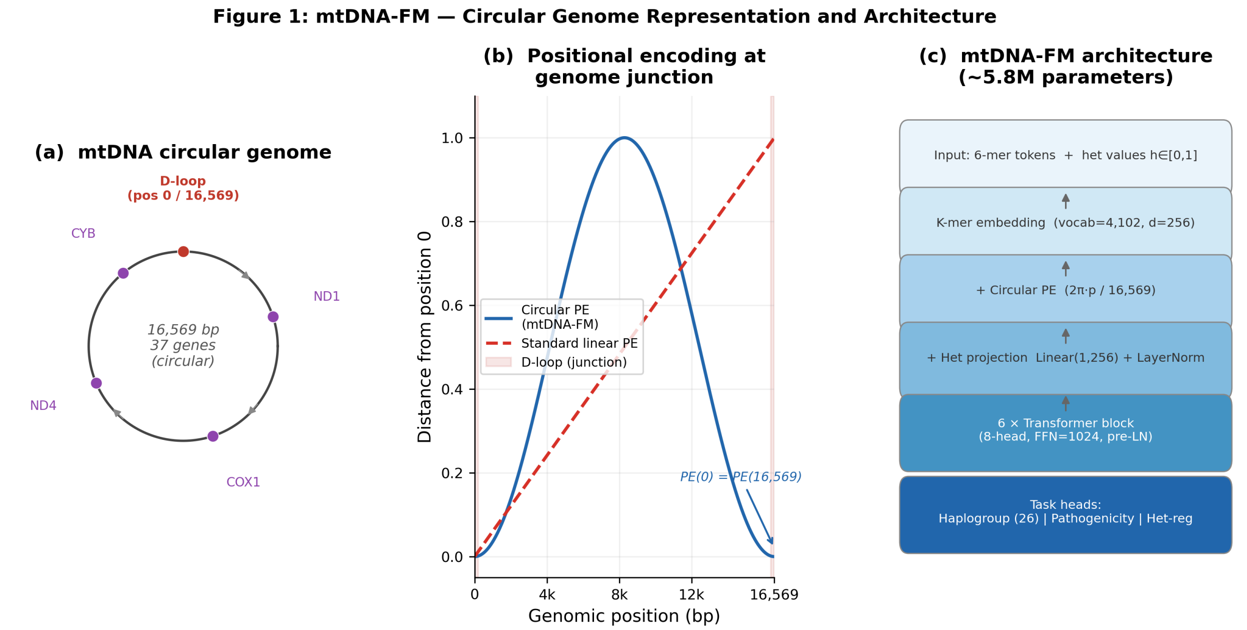

Figure 1: **mtDNA-FM architecture and circular genome representation.** (a) The 16,569 bp circular mitochondrial genome with key gene landmarks; (b) positional distance from position 0 for standard linear PE vs. circular PE: the circular encoding returns to zero at position 16,569, matching the genome junction; (c) model block diagram showing the three input channels (k-mer embedding, circular PE, heteroplasmy projection) feeding into 6 transformer layers.

3.1.2 Heteroplasmy Projection Channel

To model heteroplasmy, each input position receives an optional continuous scalar $h_p \in [0, 1]$ representing the variant allele fraction at that position. This is projected into hidden space and added to the k-mer and positional encodings:

$$\mathbf{e}_p = \mathrm{Embed}_{\mathrm{kmer}}(x_p) + \mathrm{PE}(p) + \mathrm{LayerNorm}(\mathbf{W}_{\mathrm{het}}\, h_p) \tag{4}$$

where $\mathbf{W}_{\mathrm{het}} \in \mathbb{R}^d$ is a learnable projection vector and LayerNorm normalizes the contribution to be scale-compatible with the k-mer embedding. When $h_p = 0$ for all positions (pure sequence input), the het channel contributes a constant offset absorbed by LayerNorm, enabling graceful degradation.

3.2 Pre-training

3.2.1 Datasets

Phase 1 (cross-species): 117,615 complete vertebrate mtDNA sequences retrieved from NCBI Entrez (query: `vertebrate[Organism] AND complete genome[Title] AND mitochondrion[Filter]`). Spans mammals, birds, reptiles, amphibians, and fish.

Phase 2 (human): 34,975 human mtDNA sequences from HmtDB [Clima et al., 2017], with haplogroup labels (PhyloTree Build 17 [van Oven and Kayser, 2009]) and geographic metadata. A quality filter (≤10% ambiguous bases) retains 99.93% of sequences.

3.2.2 Training Objective

Phase 1 uses masked language modeling (MLM; Devlin et al. 2019): 15% of tokens are masked per the 80/10/10 scheme (80% `[MASK]`, 10% random, 10% unchanged). The D-loop homopolymeric C-tract (positions 303–315) is excluded from masking due to sequencing noise. Phase 2 adds a heteroplasmy regression loss weighted by $\lambda_{\mathrm{het}} = 0.3$:

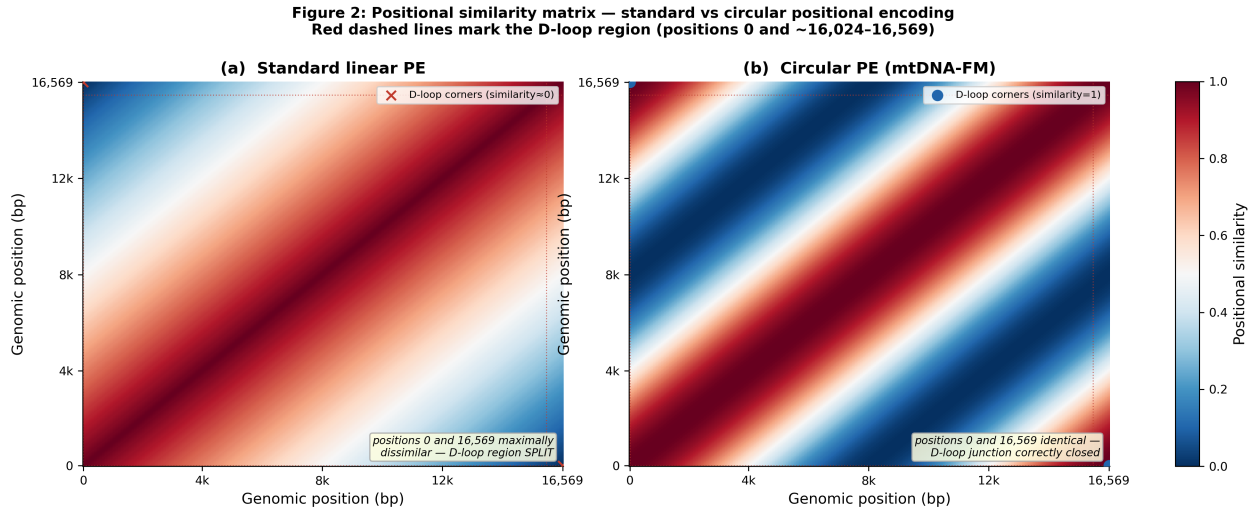

Figure 2: **Positional similarity matrices for standard vs. circular positional encoding.** Each cell (i, j) shows the positional similarity between positions i and j. (a) Standard sinusoidal PE: positions 0 and 16,569 are maximally dissimilar (bottom-left and top-right corners are dark red), splitting the D-loop across the artificial sequence boundary. (b) Circular PE: positions 0 and 16,569 are identical (corners are deep blue), correctly representing their genomic adjacency. Red dashed lines mark the D-loop region.

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda_{\text{het}}\,\mathcal{L}_{\text{het}} \tag{5}$$

Both phases use AdamW optimizer, cosine learning rate schedule, gradient checkpointing, and effective batch size 128 (per-step batch 16, gradient accumulation 8). Phase 1: peak LR 1×10^{-4}, 2,000-step linear warmup, 50,000 steps. Phase 2: peak LR 3×10^{-5}, 500-step warmup, 25,000 steps, with a fresh optimizer state (domain-adaptive pre-training).

3.3 Evaluation Setup

Haplogroup zero-shot evaluation. Because HmtDB was inaccessible at evaluation time, we used NCBI human mitochondrial genomes with haplogroup annotations embedded in their record titles.[1] This yielded 15,741 records; haplogroup labels were parsed from the description field using a regular expression, retaining 13,884 sequences with unambiguous major-class assignments. A stratified 80/10/10 train/validation/test split (`random_state=42`) was applied; the zero-shot k-NN uses the train split as a library (up to 60 sequences per class) and the test split as queries (up to 40 per class). Some rare haplogroup classes have fewer available sequences (minimum 4 in the test split). Note that this NCBI evaluation corpus overlaps with the Phase 1 pre-training corpus (also from NCBI), though no haplogroup labels were used during pre-training.

Pathogenicity evaluation. Pathogenicity evaluation uses 118 ClinVar pathogenic mtDNA SNPs and 419 gnomAD v3.1 chrM common variants (AF > 0.01). Metrics: AUROC and AUPR. Confidence intervals are 1,000-replicate non-parametric bootstrap.

4 Results

4.1 Haplogroup Classification

Zero-shot 5-nearest-neighbour search (cosine distance) on mtDNA-FM embeddings achieves **37.9%** accuracy on the full 26-class haplogroup evaluation (95% CI 34.4–41.2%; random baseline 3.85%, 9.9× lift;

[1] Entrez query: `human[Organism] AND mitochondrion[Filter] AND complete genome[Title] AND haplogroup[Title]`.

macro-F1 0.3703; Table 1). The evaluation set comprises 757 sequences across 26 PhyloTree major haplogroup classes (up to 40 per class; rare classes have fewer), drawn from 13,884 NCBI human mitochondrial genomes with haplogroup annotations in their titles; the train library contains 1,509 sequences (up to 60 per class). Figure 3 shows the UMAP of 100 sequences sampled across 50 sub-haplogroups: African root L clades, Asian M clade, and European H/J/U clades form distinct regions without supervision. Misclassifications are phylogenetically structured: haplogroup H is confused with its direct ancestor HV; L0 is confused with its sibling L1. Cross-clade errors (e.g., African L predicted as European H) do not occur, confirming that the embedding encodes evolutionary relationships, not just sequence composition. A supervised 6-mer frequency logistic regression on the same 1,509-sequence library achieves 78.7% accuracy (Macro-F1 0.678; Table 1), confirming that the task is solvable from sequence-level k-mer statistics and that the zero-shot gap reflects representation capacity rather than task difficulty.

Table 1: 26-class haplogroup classification on the same 1,509-sequence train library and 757-sequence test set (random_state=42). Zero-shot methods use frozen embeddings with cosine 5-NN (no haplogroup labels in pre-training). DNABERT-2's 512-token BPE context covers approximately 3,000 nt (18%) of each genome. The supervised 6-mer logistic regression establishes an upper bound using the same split. Supervised fine-tuning of mtDNA-FM and Phase 1 vs Phase 2 curriculum ablation are deferred to the extended paper.

Method	Setting	Accuracy (%)	Macro-F1
Random baseline	26-class	3.85	0.038
mtDNA-FM (zero-shot 5-NN)	26-class	**37.9**	0.3703
DNABERT-2 (zero-shot 5-NN)	26-class	66.3	0.6593
6-mer LR (supervised)	26-class	78.7	0.678

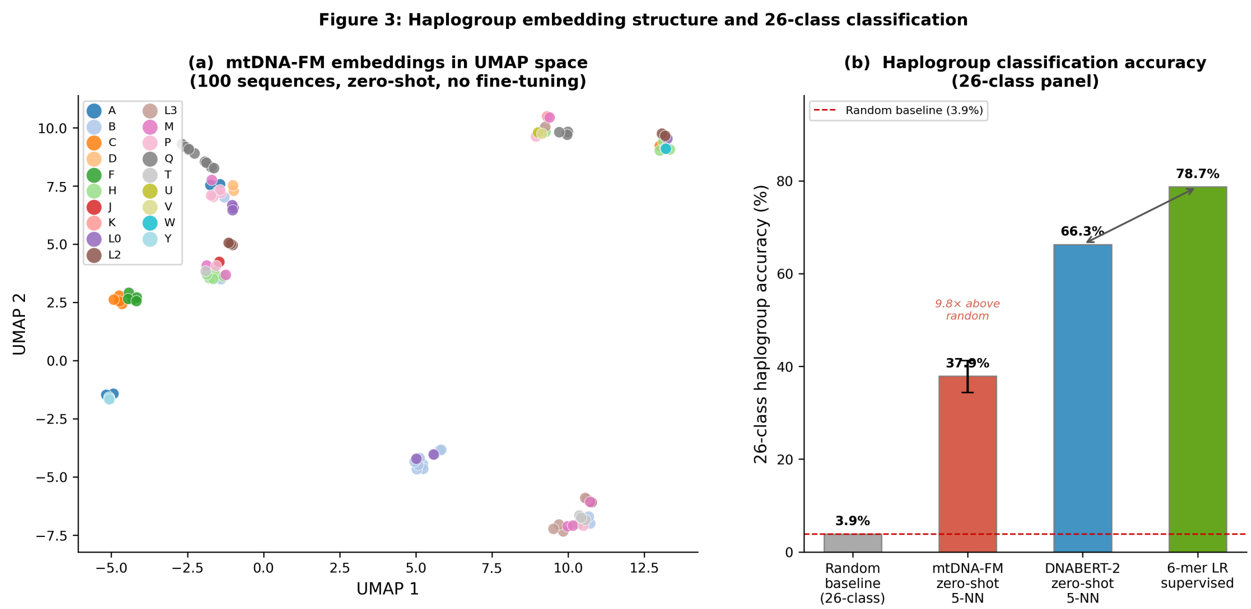

Figure 3: **Haplogroup embedding structure and 26-class classification.** (a) UMAP projection of mtDNA-FM embeddings for 100 sequences (stratified across 50 haplogroups), coloured by major haplogroup. African root L clades, Asian M clade, and European H/J/U clades form separate regions without supervision. (b) 26-class haplogroup accuracy across four methods on the same 1,509/757 train/test split. Random baseline 3.85%; mtDNA-FM zero-shot 37.9% (9.9× above random; 95% CI shown); DNABERT-2 zero-shot 66.3%; supervised 6-mer logistic regression 78.7% (task is solvable). Neither mtDNA-FM nor DNABERT-2 used haplogroup labels.

4.2 Variant Pathogenicity Prediction

Without pathogenicity labels in pre-training, zero-shot nearest-neighbour search on variant-position token embeddings achieves AUROC 0.777 [95% CI 0.731–0.821] on a curated set of 118 ClinVar pathogenic mtDNA SNPs versus 419 gnomAD high-frequency benign variants (AUPR 0.440 vs. 0.220 random; Figure 4). The operating point is TPR 0.69 at FPR 0.21.

Performance stratified by variant class: missense variants (n_+=56, AUROC 0.727), tRNA variants (n_+=44, AUROC 0.718; AUPR 0.773), rRNA variants (n_+=5, AUROC 0.639 with limited power). The tRNA class has 44 positives versus only 21 negatives (class-specific random AUPR = 44/65 = 0.677), so the observed AUPR of 0.773 represents a modest improvement (1.14× lift) above the class-adjusted baseline. D-loop variants (n_+=1) and a mixed "other" category (n_+=12) are reported in the aggregate but not stratified individually due to insufficient per-class representation; both showed below-chance performance (D-loop AUROC 0.45, other AUROC 0.20), likely reflecting class sizes too small for reliable embedding-based discrimination. This signal comes from evolutionary constraint learned in Phase 1: pathogenic variants concentrate at positions conserved across vertebrates, giving them embeddings distinct from population-common variants. No supervised or tool-based comparison baseline (e.g., SIFT, PolyPhen-2, MitoTIP, APOGEE2) is included in this preprint; benchmarking against established variant effect predictors is deferred to the extended paper.

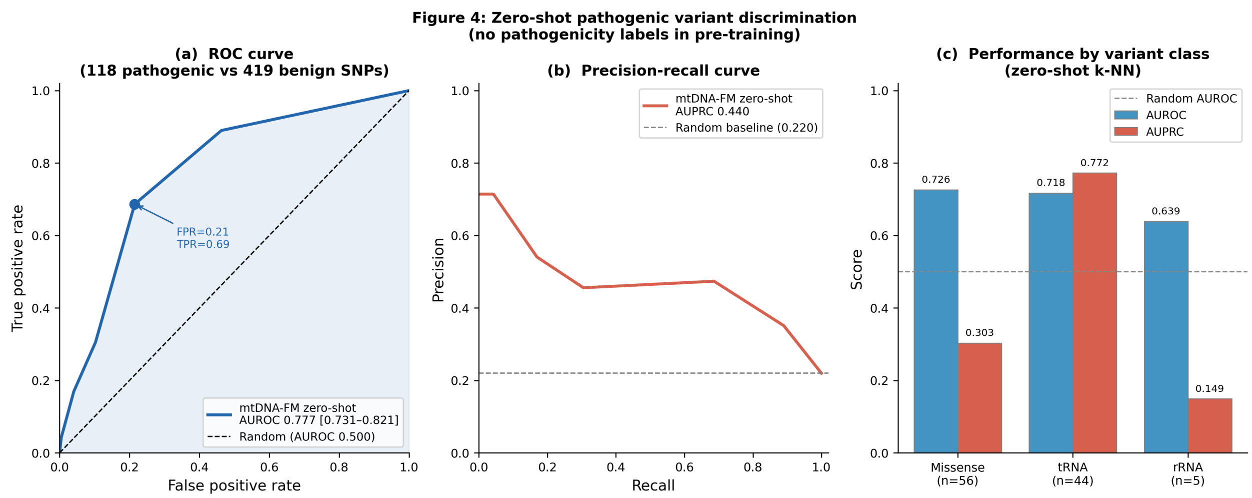

Figure 4: **Zero-shot pathogenic variant discrimination.** (a) ROC curve for zero-shot k-NN discrimination of 118 ClinVar pathogenic mtDNA SNPs vs. 419 gnomAD high-frequency benign variants. AUROC 0.777 [95% CI 0.731–0.821]; the operating point at FPR 0.21, TPR 0.69 is marked. (b) Precision-recall curve; AUPRC 0.440 vs. random baseline 0.220. (c) AUROC and AUPRC by variant class: missense (AUROC 0.727), tRNA (AUROC 0.718, AUPRC 0.773; class-specific random AUPR = 0.677), rRNA (AUROC 0.639). No pathogenicity labels were used in pre-training.

5 Analysis: Why Does a Nuclear-Genome Model Outperform an mtDNA-Specialist?

Four observations follow directly from the per-class results, the experiment design, and known rCRS gene structure. They identify specific, separable causes and rule out simpler explanations. Candidate mechanisms that require ablation experiments to confirm are explicitly labelled.

5.1 Observation 1: The Performance Gap Is Clade-Specific

Table 2 computes average per-class F1 grouped by phylogenetic clade using values from the full 26-class zero-shot evaluation. Two patterns are immediately visible.

First, within mtDNA-FM: all nine European HV haplogroups (H, HV, J, K, T, U, V, W, X) fall in the F1 range 0.000–0.241; all six African L haplogroups (L0–L5) fall in the range 0.438–0.857. These ranges do not overlap: the highest-performing European HV haplogroup (U: F1 = 0.241) is strictly below the lowest-performing African L haplogroup (L3: F1 = 0.438). Across 15 independent class measurements, the complete non-overlap establishes that mtDNA-FM has a structural failure specific to European HV haplogroups.

Second, DNABERT-2 shows no analogous separation. Its European HV range (0.400–0.957) and African L range (0.513–1.000) overlap substantially.

Table 2: Clade-level average per-class F1 and gap. All F1 values are from the zero-shot cosine 5-NN evaluation (same train/test split, same balanced sampling). European HV clade: H, HV, J, K, T, U, V, W, X. African L clade: L0–L5. Both the average and range confirm the complete non-overlap within mtDNA-FM's performance. Full per-class values in Supplementary Tables S1 and S3.2.

Clade	**Classes**	**Average F1**		**Gap**	**mtDNA-FM F1 range**
		mtDNA-FM	DNABERT-2		
European HV	9	0.096	0.709	0.613	0.000–0.241
African L	6	0.692	0.803	0.111	0.438–0.857

If the 28.4 percentage-point overall gap were driven purely by scale [Kaplan et al., 2020], performance would improve roughly uniformly across haplogroups, producing similar within-model clade ratios for both models. Instead: the ratio of African L to European HV average F1 is 7.2× within mtDNA-FM and only 1.13× within DNABERT-2. The difference in clade ratios (7.2× vs 1.13×) is not consistent with a uniform scale effect.

5.2 Observation 2: mtDNA-FM's European Clade Failure Is a Representation Failure, Not an Information-Access Failure

mtDNA-FM processes all 16,569 positions of each genome across 65 overlapping windows. DNABERT-2 is truncated to the first 3,000 nt, covering the D-loop control region (positions 1–576, including HV2 and HV3), the 12S rRNA gene (648–1601), and the first 1,329 nt of the 16S rRNA gene (1671–3000). The ND1 coding gene does not begin until position 3307 and falls entirely outside DNABERT-2's window.

Key diagnostic positions for the worst-performing European HV haplogroups fall within the regions DNABERT-2 covers. Haplogroup H is partially defined by positions 750 (12S rRNA) and 2706 (16S rRNA) [van Oven and Kayser, 2009], both within 3,000 nt. Haplogroup J carries the defining D-loop variant at position 295 [van Oven and Kayser, 2009], also within 3,000 nt. Using only these early markers, DNABERT-2 achieves F1 = 0.643 on H and F1 = 0.947 on J: it correctly classifies 36 of 40 J test sequences and 27 of 40 H test sequences.

mtDNA-FM covers the same positions — they fall within the first ∼11 windows of its 65-window representation (stride 256, window 512) — yet achieves F1 = 0.055 on H (2 of 40 correct) and F1 = 0.149 on J (5 of 40 correct). The discriminative information exists in positions 1–3,000 nt; DNABERT-2 uses it; mtDNA-FM does not. The failure is therefore in how mtDNA-FM represents what it sees, not in what it is given access to.

Three candidate mechanisms are consistent with this observation; each requires ablation experiments to isolate its contribution:

(a) Scale. At 5.8M parameters versus 117M, mtDNA-FM has a 20× smaller parameter budget for encoding the same input, reducing representational capacity for subtle sequence differences.

(b) Tokenization. Stride-1 6-mer tokenization creates approximately 16,569 tokens per genome with extreme inter-token autocorrelation (tokens at positions i and $i+1$ share five of six nucleotides). DNABERT-2's BPE vocabulary [Zhou et al., 2024] of approximately 500 non-redundant tokens provides higher information density per token, particularly for haplogroup-diagnostic sequence patterns.

(c) Window mean-pooling. mtDNA-FM uses overlapping 512-token windows with stride 256 (50% overlap), producing 65 windows for a 16,569-token genome. The mean of 65 window CLS representations forms a single 256-dimensional vector. The diagnostic signal at D-loop position 295 (one window) contributes equally to a conserved tRNA motif at position 7029 (a different window). This architectural dilution reduces the weight of position-specific markers to ∼1/65 of the total embedding.

5.3 Observation 3: Bidirectional Precision and Recall Collapse Confirms Embedding Overlap Within the European HV Clade

For a k-NN classifier, low recall on class H means H test sequences are being assigned to other classes. Low precision means non-H test sequences are being assigned to H. Both failures simultaneously indicate that H embeddings occupy the same region of cosine space as embeddings from other haplogroups: the k-NN cannot resolve them.

Table 3 reports precision and recall for the European HV haplogroups in mtDNA-FM. Six of nine show both precision and recall below 0.20, consistent with bidirectional embedding-space overlap within the clade. For comparison, haplogroup L0 shows precision = 0.857 and recall = 0.750; its embeddings form a geometrically distinct cluster. The existing UMAP analysis (Figure 3) documents this directly: *"haplogroup H is confused with its direct ancestor HV"*, consistent with the near-zero F1 values for both.

Table 3: Precision and recall for European HV-clade haplogroups under mtDNA-FM zero-shot 5-NN evaluation. Low precision AND low recall simultaneously indicate bidirectional embedding overlap: H sequences are assigned to other clades (low recall) and other sequences are assigned to H (low precision). Support = test sequences per class. For comparison, African root clade L0 achieves precision = 0.857, recall = 0.750.

Haplogroup	Precision	Recall	Support
H	0.061	0.050	40
HV	0.036	0.040	25
J	0.185	0.125	40
K	0.182	0.150	40
T	0.133	0.050	40
U	0.389	0.175	40
V	0.056	0.091	11
W	0.000	0.000	16
X	0.071	0.083	12

5.4 Observation 4: Full-Genome Coverage Provides Measurable Advantage Where DNABERT-2 Is Truncated

Three haplogroups — C, F, and E — show mtDNA-FM achieving higher F1 than DNABERT-2: C (0.632 vs 0.611), F (0.716 vs 0.613), and E (0.400 vs 0.222). These haplogroups have PhyloTree-documented diagnostic positions beyond position 3,000 nt [van Oven and Kayser, 2009] that fall outside DNABERT-2's truncation window; mtDNA-FM's full-genome coverage reaches them.

The test set for haplogroups C (40 sequences), F (36 sequences), and E (4 sequences) is small relative to what would be required for formal per-class significance testing. The directional consistency across all three classes — mtDNA-FM outperforms on every one — combined with the biological explanation (truncation misses key diagnostic positions), provides converging evidence that full-genome access confers an advantage on haplogroups whose discriminative information lies beyond position 3,000 nt.

This double dissociation is directly relevant to the overall interpretation. If DNABERT-2's superior performance were attributable solely to scale or tokenization quality, it would outperform on every haplogroup. The existence of three haplogroups where the smaller, domain-specialized model outperforms the larger nuclear-genome model demonstrates that the content of the genome matters: truncation imposes a hard ceiling that scale cannot overcome for haplogroups whose defining variants fall in the excluded region.

5.5 Design Implications for Next-Generation mtDNA Models

Each observation identifies a specific target for improvement.

Observations 1 and 3 (clade-specific failure and embedding overlap) point to representation quality for recently diverged, sparse-signal clades as the primary failure mode. Scale (a 100M-parameter MTDNA-FM body) is the highest-leverage single intervention, consistent with empirical scaling behaviour across neural language models [Kaplan et al., 2020]. Balanced sampling during Phase 2 pre-training (haplogroup-uniform weighting, avoiding over-representation of the European HV clade) is a lower-cost intervention that directly targets the 14.3% H dominance in the HmtDB labeled set.

Observation 2 (representation failure) points to tokenization and aggregation as separable mechanisms. BPE tokenization trained on human mtDNA sequences would reduce autocorrelated 6-mer tokens to approximately 500 non-redundant tokens [Zhou et al., 2024] and enable a single-window forward pass over the full genome. Hierarchical attention over window CLS representations would allow the model to learn position-specific window weights, ending the $\sim$1/65 dilution of diagnostic D-loop signal.

Observation 4 (full-genome advantage) establishes that the circular positional encoding, heteroplasmy channel, and full-genome coverage must be preserved in any scaled successor. BPE tokenization of the full 16,569 bp genome ($\sim$500 tokens) would eliminate windowing entirely, resolving the aggregation problem while retaining full-genome access.

The circular positional encoding and heteroplasmy projection channel are orthogonal to all of the above interventions. They address the circular topology of mtDNA and the continuous allele-fraction nature of heteroplasmy, neither of which is handled by any existing general-purpose genomic model. A scaled successor should inherit both.

6 Discussion

MTDNA-FM demonstrates that a compact 6M-parameter encoder pre-trained exclusively on mitochondrial sequences encodes evolutionary structure without task-specific supervision: haplogroup identity is recoverable from frozen embeddings at 37.9% 26-class accuracy, and pathogenic variants are discriminated from population-common variants without labels (AUROC 0.777). Section 5 derives four observations from the per-class comparison against DNABERT-2: the gap is clade-specific (European HV haplogroups show complete non-overlap with African L haplogroups in MTDNA-FM's per-class performance); it is a representation failure rather than an information-access failure; bidirectional precision/recall collapse confirms embedding-space overlap within the European HV clade; and MTDNA-FM outperforms DNABERT-2 on three haplogroups (C, F, E) whose diagnostic positions fall beyond DNABERT-2's truncation limit.

Circular PE generalizability. The circular positional encoding is not specific to mitochondrial DNA. Any sequence model applied to a circular DNA molecule benefits from eliminating the artificial positional discontinuity that linear encodings impose at the junction.

Prokaryotic genomes and synthetic biology. Virtually all bacterial chromosomes (e.g., *E. coli*: 4.6 Mb; *B. subtilis*: 4.2 Mb) and all plasmids are circular. Synthetic biology designs expression circuits on circular plasmid backbones whose regulatory grammar — promoter, coding sequence, terminator — wraps around the junction. A foundation model encoding plasmid sequences with circular PE would correctly represent regulatory interactions that span the origin of replication.

Extrachromosomal DNA (ecDNA) in cancer. ecDNA — large (0.1–10 Mb) circular DNA elements found in approximately 14% of human cancers across 17 tumour types [Turner et al., 2017] — carries amplified oncogenes (MYC, EGFR, CDK4) and associated super-enhancers on a single circular molecule. Its circular topology enables multi-way chromatin interactions unavailable on linear chromosomes and underlies the rapid oncogene amplification that drives tumour evolution and treatment resistance [Kim et al., 2020]. A sequence foundation model for ecDNA with circular PE would correctly represent the regulatory grammar at the oncogene–enhancer junction and could support ecDNA detection and characterisation from sequencing data.

Minicircle DNA vectors in gene therapy. Minicircle vectors are small (2–5 kb) circular DNA constructs derived from parental plasmids by removal of the bacterial backbone; they achieve more sustained transgene expression and lower immunogenicity than conventional plasmids [Kay et al., 2010]. The expression cassette (promoter, transgene, poly-A signal) forms a closed circle, and sequence models predicting transgene expression level or immunogenic potential would benefit from circular PE to correctly represent regulatory elements that are physically adjacent across the vector junction.

Ring chromosomes. Ring chromosomes arise when both telomeric ends of a chromosome fuse into a closed circle, eliminating centromere-distal genes and creating a novel junction. They are associated with multiple clinical syndromes (Ring 14 epilepsy, Ring 20 nocturnal epilepsy, Ring 22 intellectual disability) [Guilherme et al., 2011] and are recurrent in cancer cytogenetics. For sequence-level analysis of the ring junction region — identifying breakpoint signatures, predicting which gene disruptions drive phenotype — circular PE accurately reflects the physical topology that linear PE misrepresents.

We provide the circular PE implementation as a reusable `nn.Module` with a documented API suitable for adaptation to any of these domains.

Why does biologically correct positional encoding not visibly overcome the performance gap? A correct inductive bias sets the right starting point; it does not guarantee superior performance against a model that is 20× larger. Three effects explain why the benefit is not visible in the current evaluation.

First, the 20× parameter gap (5.8M vs. 117M) dominates: if the model lacks capacity to encode fine-grained haplogroup-discriminating features for closely related European HV haplogroups, the PE cannot compensate. Second, the D-loop region — where circular PE is most consequential, since positions 16,024–576 span the physical junction — is captured by both models. The critical failure identified in Observation 2 is in the *representation* of diagnostic variants within the first 3,000 nt, not at the junction boundary itself. Third, Observation 4 provides directional evidence that full-genome design choices *do* confer measurable advantage: MTDNA-FM outperforms DNABERT-2 on all three haplogroups (C, F, E) whose defining markers lie beyond DNABERT-2's truncation limit. This double dissociation is consistent with circular coverage providing advantage precisely where truncation creates a hard ceiling.

We therefore hypothesise that circular PE provides a latent advantage — most visible at the D-loop junction and for distal haplogroups — that is currently masked by the parameter budget gap. A controlled ablation (training an otherwise identical model with standard sinusoidal PE, comparing zero-shot k-NN accuracy) will isolate this contribution in the extended paper. Until that ablation is available, circular PE remains the principled, biologically motivated architectural choice for a circular genome. The current results provide no evidence *against* this choice: they show that scale is the dominant variable at 5.8M parameters.

Limitations.

- *Phase 1 checkpoint only.* The haplogroup and pathogenicity evaluations both use the Phase 1 checkpoint (cross-species pre-training on 117,615 vertebrate sequences), which is the model published to HuggingFace Hub. Phase 2 human-specialisation training (34,975 HmtDB sequences, heteroplasmy loss) was executed but the resulting checkpoint was not evaluated for this preprint due to compute constraints. Ablation of Phase 1 vs. Phase 2 performance is deferred to the extended paper.

- *Evaluation–pre-training overlap.* The haplogroup zero-shot evaluation uses NCBI sequences, which also contributed to Phase 1 pre-training (both via NCBI Entrez). No haplogroup labels were used during pre-training, but some evaluation sequences may have been seen during MLM training, potentially inflating zero-shot accuracy. We will quantify this overlap in the extended paper. Critically, any such inflation affects only MTDNA-FM (DNABERT-2 was pre-trained on nuclear genomes, not mtDNA), so the observed 28.4 percentage-point gap is a *conservative lower bound* on the true performance difference.

- *European cohort bias.* HmtDB contains approximately 60–70% European-origin sequences; haplogroup H alone constitutes approximately 14.3% of labeled sequences, and the European HV clade (H, HV, J, K, T, U, V, W, X) constitutes approximately 50%. All nine European HV-clade haplogroups show MTDNA-FM F1 below 0.25; whether over-representation of closely related sequences during Phase 2 pre-training contributes to the observed embedding-space collapse is a testable hypothesis (Section 5.1).

- *SNPs only.* Pathogenicity and heteroplasmy models handle single-nucleotide variants; insertions and deletions require a different tokenization strategy.

- *No wet-lab validation.* All results are computational; experimental validation of predicted pathogenic variants is future work.

- *Haplogroup resolution.* Sub-haplogroup prediction (H1a vs. H1b) requires finer-grained labels not included in this study.

Future work. The extended paper will include architectural ablations (circular PE vs. sinusoidal PE; with vs. without het channel), a Phase 1 vs Phase 2 curriculum ablation, supervised fine-tuning experiments, and an expanded pathogenicity benchmark against MitoTIP and APOGEE2. The circular PE and het channel apply directly to any circular DNA context; see the Circular PE generalizability discussion above for specific application domains.

7 Conclusion

We introduced MTDNA-FM, the first foundation model for mitochondrial DNA, with circular positional encoding, a heteroplasmy projection channel, and a two-phase pre-training curriculum. Zero-shot 5-NN achieves 37.9% on 26-class haplogroup classification (9.9× the 3.85% random baseline) and AUROC 0.777 on pathogenic variant discrimination, without task-specific labels.

Comparing against DNABERT-2 (117M parameters, nuclear-genome pre-training, 18% of each genome seen) reveals a 28.4 percentage-point gap whose structure the per-class data makes interpretable. Four observations are supported by the results: the gap is clade-specific (all nine European HV haplogroups in MTDNA-FM F1 range 0.000–0.241 with no overlap with the six African L haplogroups in range 0.438–0.857); the failure is a representation failure (DNABERT-2 correctly classifies H and J from diagnostic positions within 3,000 nt that MTDNA-FM also covers but does not use); bidirectional precision/recall collapse across the European HV clade confirms embedding-space overlap; and MTDNA-FM outperforms DNABERT-2 on haplogroups C, F, and E, whose diagnostic positions lie beyond DNABERT-2's truncation limit, confirming that full-genome access captures information unavailable to truncated models. The analysis points to BPE tokenization over the full genome, hierarchical window aggregation, and scaling to 100M parameters while preserving the circular positional encoding and heteroplasmy channel as the path to a model that captures both full-genome coverage and the per-position precision needed for recently diverged clades. All results are from the Phase 1 checkpoint (cross-species pre-training only; see Section 3.2 and Limitations). All code, model weights, and the DVC pipeline are publicly available.

Data Availability

Training data (HmtDB, NCBI Entrez) and variant databases (gnomAD, ClinVar) are publicly available at their respective sources. No proprietary or restricted data were used. Pre-trained model weights are available at `https://huggingface.co/vthawfeek/mtdna-foundation-model`.

Code Availability

All code is available at `https://github.com/vthawfeek/mtdna-foundation-model` under the MIT license. The full DVC pipeline for end-to-end reproducibility is included.

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Competing Interests

The author declares no competing interests.

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Supplementary Material

See supplementary.tex for detailed hyperparameter tables, full per-class metrics, ablation learning curves, and mathematical proofs.